Algebraic Topology

Shing Tak Lam

Michaelmas 2023*

Contents

1 Introduction **1**
1.1 Co(chain) complexes 2

Lecture 1

1 Introduction

Let X, Y be topological spaces, $f, g : X \to Y$ be continuous functions. Note we will use the terminology 'map := continuous function'. We say that f *is homotopic to* g, written $f \simeq g$, if there exists a map $F : X \times I \to Y$, such that

$$F|_{X\times\{0\}} = f \quad \textit{and} \quad F|_{X\times\{1\}} = g$$

Here $I = [0, 1]$ carries its Euclidean topology. An exercise (sheet 1) is that this defines an equivalence relation on the set of maps $X \to Y$.

Definition 1.1 (homotopy equivalence)
We say that $f : X \to Y$ is a *homotopy equivalence* if there exists a function $g : Y \to X$ such that $f \circ g \simeq \mathrm{id}_Y$ and $g \circ f \simeq \mathrm{id}_X$. We say that g is a *homotopy inverse* to f.

The same exercise on sheet 1 shows that homotopy equivalence is an equivalence relation on topological spaces.

Example 1.2
1. If $f : X \to Y$ is a homeomorphism, then f is also a homotopy equivalence.
2. $i : \{0\} \hookrightarrow \mathbb{R}^n$ is a homotopy equivance.
3. The inclusion $i : S^{n-1} \hookrightarrow \mathbb{R}^n \setminus 0$ is a homotopy equivalence.

Definition 1.3
A space X is *contractible* if it is homotopy equivalent to a point.

Idea of Algebraic Topology: Study spaces up to homotopy equivalence. Broadly, we are interested in "connectivity properties" of topological spaces.

Example 1.4
1. We say a space X is *path-connected* if any two maps $\{*\} \to X$ are homotopic. For example, $\mathbb{R}$ is path connected, $\mathbb{R} \setminus \{0\}$ is not. A corollary of this is the intermediate value theorem.
2. We say a path connected space X is *simply connected* if every map $f : S^1 \to X$ is homotopic to a constant map. Equivalently, every two maps $S^1 \to X$ are homotopic. Or equivalently (again) every continuous map $S^1 \to X$ extends to a continuous map $D^2 \to X$. For example, $\mathbb{R}^2$ is simply

*Based on lectures by Ivan Smith. Last updated October 9, 2023.

connected, by $\mathbb{R}^2 \setminus \{0\}$ is not. If $\gamma : S^1 \to \mathbb{R}^2 \setminus \{0\}$ is continuous, it has a *winding number*, or *degree* $\deg(\gamma) \in \mathbb{Z}$, such that

- It is invariant under homotopy,
- If $\gamma_n(t) = \exp(2n\pi it)$, then $\deg(\gamma_n) = n$

In particular, taking $n = 0$ gives the constant map. Therefore, a constant map has degree 0. A corollary of this is the fundamental theorem of algebra.

3. We say a path connected space X is *k-connected*, if for all $i \leq k$, every map $S^i \to X$ is homotopic to a constant map. In this case, $\mathbb{R}^n$ is $(n-1)$-connected, but $\mathbb{R}^n \setminus 0$ is not. More precisely, maps from $S^{n-1} \to \mathbb{R}^n \setminus 0$ have a degree $\deg(f) \in Z$, such that degree is a homotopy invariant, and the degree of the inclusion map is 1, and the degree of the constant map is 0. A corollary of this is the Brouwer fixed point theorem.

Lecture 2

1.1 Co(chain) complexes

We will define invariants of topological spaces in two steps:

(a) We associate to X a (co)chain complex,

(b) We take the (co)homology of the complex

The topology enters in step (a), whereas (b) is just homological algebra.

Definition 1.5 (chain complex)

A *chain complex*@ (C_*, ∂) consists of a sequence $\{C_i\}_{i\in\mathbb{Z}}$ of abelian groups, along with *boundary homomorphisms* $\partial_n : C_n \to C_{n-1}$ for all n, satisfying $\partial_n \circ \partial_{n+1} = 0$. We also write this conditions as $\partial^2 = 0$.

Definition 1.6 (homology group)

The *i-th homology group* of a chain complex (C_*, ∂) is

$$\mathrm{H}_i(C_*, \partial) = \frac{\ker(\partial_i : C_i \to C_{i-1})}{\mathrm{im}(\partial_{i+1} : C_{i+1} \to C_i)}$$

We write

$$\mathrm{H}_*(C_*, \partial) = \bigoplus_{i\in\mathbb{Z}} \mathrm{H}_i(C_*, \partial)$$

Definition 1.7 (cochain complex)

A *cochain complex* (C^*, ∂) consists of a sequence $\{C^i\}_{i\in\mathbb{Z}}$ of abelian groups, along with *boundary homomorphisms* $\partial^n : C^n \to C^{n+1}$ for all n, satisfying $\partial^{n+1} \circ \partial^n = 0$. We also write this conditions as $\partial^2 = 0$.

Definition 1.8 (cohomology group)

The *i-th cohomology group* of a cochain complex (C^*, ∂) is

$$\mathrm{H}^i(C^*, \partial) = \frac{\ker(\partial^i : C^i \to C^{i+1})}{\mathrm{im}(\partial^{i-1} : C^{i-1} \to C^i)}$$

We write

$$H^*(C^*, \partial) = \bigoplus_{i \in \mathbb{Z}} H_i(C^*, \partial)$$

Notation 1.9. Elements of $\ker(\partial)$ are called *cycles* (in a chain complex), *cocycles* (in a cochain complex). Elements of $\mathrm{im}(\partial)$ are called *boundaries* (and *coboundary* resp.).

Elements of H_* are called *homology classes*, and elements of H^* are called *cohomology classes*.

We will call ∂ the *differential*.

Definition 1.10 (chain map)

Given chain complex (C_*, ∂), (D_*, ∂), a *chain map* $f : C_* \to D_*$ comprises group homomorphism $f_i : C_i \to D_i$, such that

$$\begin{array}{ccc} C_i & \xrightarrow{f_i} & D_i \\ \downarrow{\scriptstyle \partial} & & \downarrow{\scriptstyle \partial} \\ C_{i-1} & \xrightarrow[f_{i-1}]{} & D_{i-1} \end{array}$$

commutes.

Lemma 1.11. A chain map $f : C_* \to D_*$ induces homomorphisms $f_* : H_i(C_*) \to H_i(D_*)$.

Proof. Let $a \in H_i(C_*)$, and we can choose a cycle $\alpha \in C_i(C_*)$, with $\partial\alpha = 0$. In this case,

$$\partial f_i(\alpha) = f_{i-1}(\partial\alpha) = f_{i-1}(0) = 0$$

So $f_i(\alpha) \in D_i$ is a cycle. We set $f_*(a) = [f_i(\alpha)] \in H_i(D_*)$. Next, we need to show that this is independent of choices. Suppose $a = [\alpha] = [\alpha']$. Then $\alpha - \alpha' \in \mathrm{im}(\partial)$. Therefore, we can write $\alpha - \alpha' = \partial\tau$. In this case,

$$f_i(\alpha) - f_i(\alpha') = f_i(\partial\tau) = \partial f_{i+1}(\tau) \in \mathrm{im}(\partial)$$

So we conclude $[f_i(\alpha)] = [f_i(\alpha')] \in H_i(D_*)$. It is clear that f_* defines a homomorphism. □

Correspondingly, we can define *cochain maps* $C^* \to D^*$, that is, $f^i : C^i \to D^i$ such that

$$\begin{array}{ccc} C^{i+1} & \xrightarrow{f^{i+1}} & D^{i+1} \\ \uparrow{\scriptstyle \partial} & & \uparrow{\scriptstyle \partial} \\ C^i & \xrightarrow[f^i]{} & D^i \end{array}$$

commutes.

Exercise: This construction is functorial. That is, $(\mathrm{id}_{C_*})_* = \mathrm{id}_{H_*}$, and if we have $f : C_* \to D_*, g : D_* \to E_*$, then $(g \circ f)_* = g_* \circ f_*$.

Goal: To associate to a topological space X (co)chain complex $C_*(X)$ and $C^*(X)$. The definition works for any X, but the theories are better behaved for "nicer" X.

We will develop singular (co)homology.

Definition 1.12 (standard simplex, face)

The *standard simplex* is

$$\Delta^n = \left\{ (t_0, \ldots, t_n) \in \mathbb{R}^{n+1} \mid t_i \geq 0, \sum_i t_i = 1 \right\}$$

The i-th face Δ^n is

$$\Delta^n_i = \Delta^n \cap \{t_i = 0\}$$

and we have a canonical homeomorphism

$$\begin{aligned} \delta_i : \Delta^{n-1} &\to \Delta^{n-1}_i \\ (t_0, \dots, t_{n-1}) &\mapsto (t_0, \dots, t_{i-1}, 0, t_i, \dots, t_{n-1}) \end{aligned}$$

Definition 1.13 (singular simplex, singular chain complex)

If X is a topological space, a *singular n-simplex* in X is a continuous map $\sigma : \Delta^n \to X$.

The *singular chain complex* is given by the the free abelian groups

$$C_i(X) = \bigoplus_{\sigma \text{ singular } n\text{-simplex}} \mathbb{Z}\sigma$$

We define the boundary map $C_i(X) \to C_{i-1}(X)$ via

$$\partial\sigma = \sum_{j=0}^{i} (-1)^j \sigma|_{\Delta^i_j}$$

where

$$\sigma|_{\Delta^i_j} := \sigma \circ \delta_j : \Delta^{n-1} \to X$$

Note if $\{v_i\}_{i=0}^n$ are $n+1$ ordered points in $\mathbb{R}^{n+1}$, and if $\{v_i - v_0\}_{i=1}^n$ are linearly independent, then the convex hull of $v_0, \dots, v_n$ is an n-simplex, which we will call

$$[v_0, \dots, v_n]$$

which is given by the map

$$\begin{aligned} \Delta^n &\to \mathbb{R}^{n+1} \\ t &\mapsto \sum_i t_i v_i \end{aligned}$$

We orient the edges of the standard simplex (and thus any simplex $[v_0, \dots, v_n]$) by saying that $v_i < v_j$ if $i < j$.

More concretely, for $n = 2$, we have the oriented edges $v_0 \to v_1, v_0 \to v_2, v_1 \to v_2$.

Lemma 1.14. $\partial^2 = 0$.

Proof. If σ is an n-simplex, defined on $[v_0, \dots, v_n]$, then

$$\partial\sigma = \sum_j (-1)^j \sigma|_{[v_0, \dots, \widehat{v_j}, \dots, v_n]}$$

and so

$$\partial(\partial\sigma) = \sum_{i<j} (-1)^i (-1)^j \sigma|_{[v_0, \dots, \widehat{v_i}, \dots, \widehat{v_j}, \dots, v_n]} + \sum_{i>j} (-1)^{i+1} - 1^j \sigma|_{[v_0, \dots, \widehat{v_j}, \dots, \widehat{v_i}, \dots, v_n]} = 0$$

since the signs cancel. □

Definition 1.15

The *singular homology groups of X*, $H_i(X)$, are the homology groups of the singular chain complex.

Remark 1.16. Since all we used is the topology on X, $H_i(X)$ is clearly a topological (read homeomorphism) invariant of X.

Moreover, if $f : X \to Y$ is continuous, $\sigma : \Delta^n \to X$ is an n-simplex on X, then $f \circ \sigma$ is an n-simplex on Y. The fact that

$$(f \circ \sigma) \circ \delta_j = f \circ (\sigma \circ \delta_j)$$

shows that f induces a chain map.

Index

boundary, 3
boundary homomorphism, 2

chain complex, 2
chain map, 3
coboundary, 3
cochain complex, 2
cochain maps, 3
cocycle, 3
cohomology class, 3
cohomology group, 2
k-connected, 2
contractible, 1
cycle, 3

differential, 3

homology class, 3
homology group, 2
homotopic, 1
homotopy equivalence, 1

path connected, 1

simply connected, 1
singular chain complex, 4
singular homology, 4
singular simplex, 4
standard simplex, 3